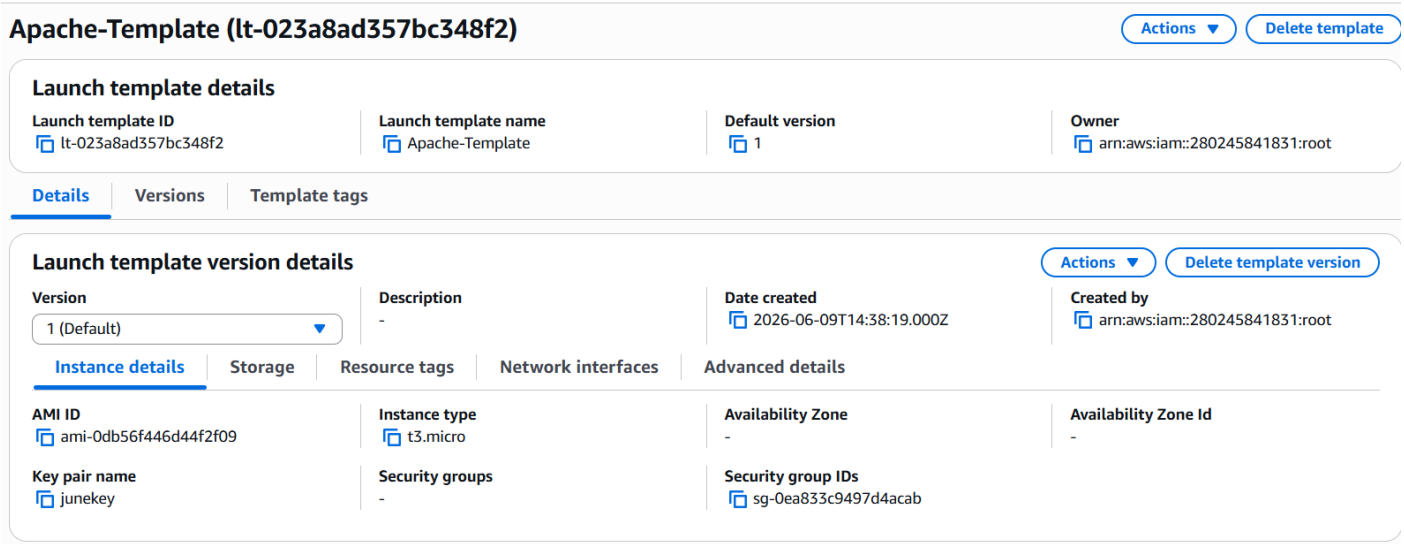
AWS AUTOSCALING

AWS Auto Scaling is in high demand today because modern businesses rely on cloud systems that must handle changing workloads efficiently. Applications such as e-commerce, banking, streaming, and enterprise platforms experience unpredictable traffic. Auto Scaling automatically increases or decreases server capacity based on real-time demand, ensuring high performance, availability, and cost optimization.

Companies use Auto Scaling to prevent downtime, improve customer experience, and reduce unnecessary cloud expenses. Since AWS is widely adopted across industries, employers expect cloud and DevOps professionals to understand Auto Scaling as part of production-ready infrastructure. Learning Auto Scaling provides practical, job-relevant skills and improves employability in cloud technology roles.

Task 1: Launch template creation

Step 1: Create a Launch template

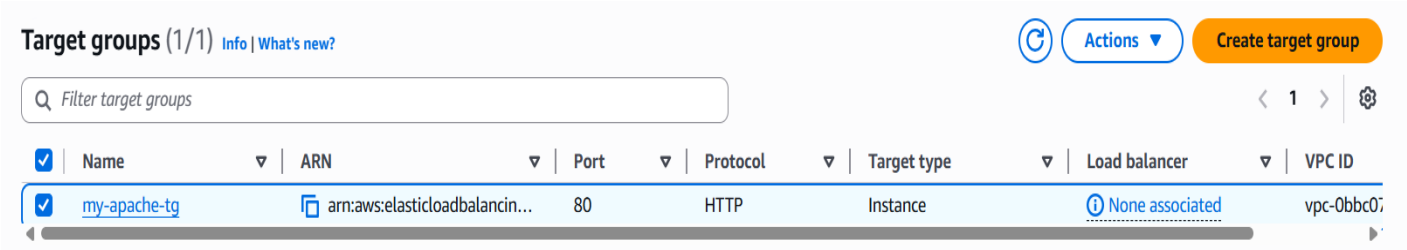


Step 2: Verify that the instance launches successfully and can be accessed using its public IP address.

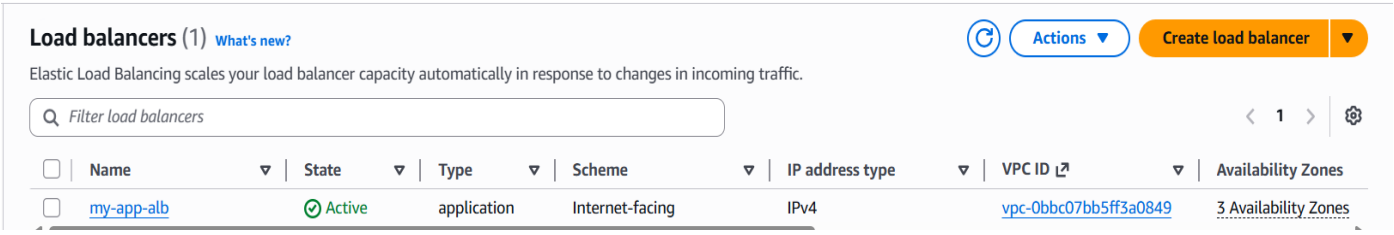
Apache Installed Automatically via Launch Template

Task 2: Auto scaling group setup

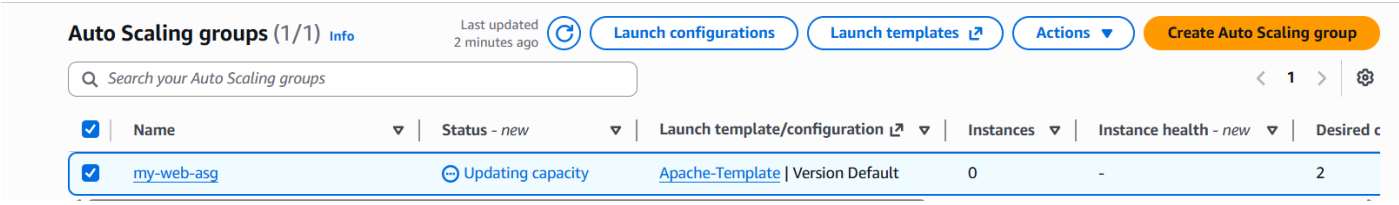
Step 1: Create Target Group



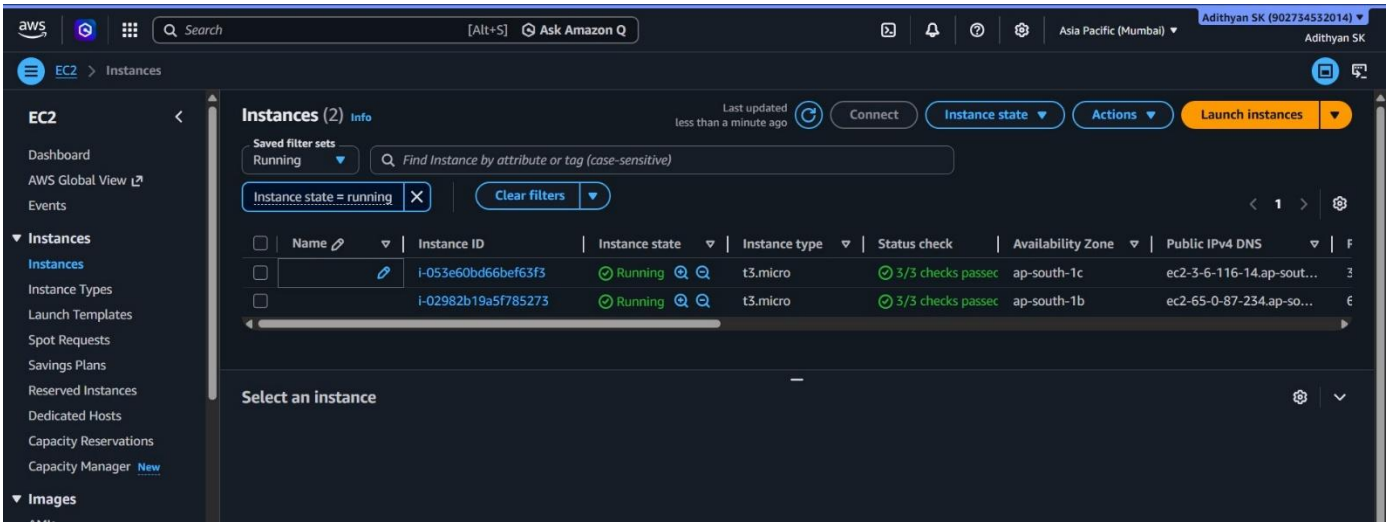
Step 2: Create Load Balancer



Step 3: Create Auto Scaling Group

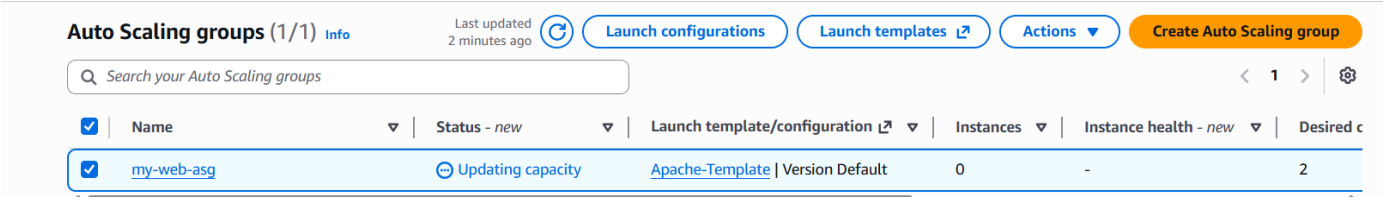


A minimum of 2 instances are launched automatically by the Auto Scaling Group

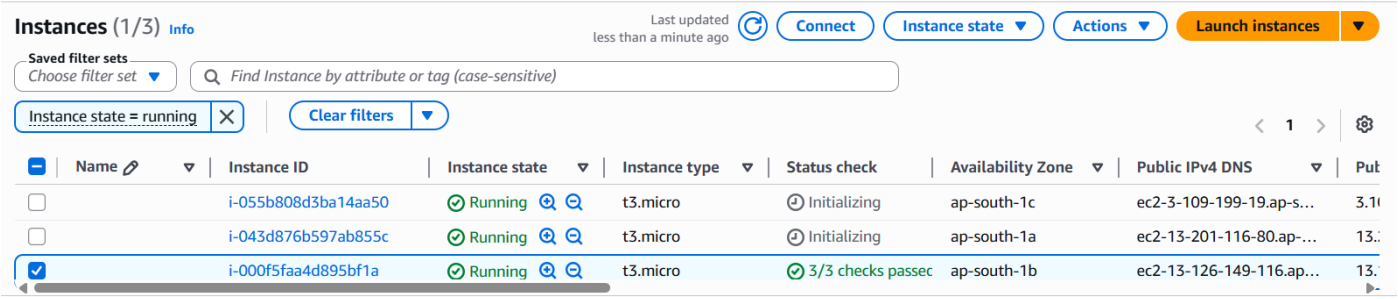


Task 3: Application Validation

Step 1: Copy Load Balancer DNS name



Instances launched automatically after attaching load balancer to autoscaling



Step 2: Open the Load Balancer DNS name in a browser and verify that the application is accessible. Refresh the page multiple times to observe different instance IDs, demonstrating traffic distribution by the Load Balancer.

Task 4: Scaling Policy Configuration

Configure a CPU-based automatic scaling policy.

Auto Scaling group: my-web-asg

Enabled or disabled
Enabled

Execute policy when
As required to maintain Average CPU utilization at 50

Take the action
Add or remove capacity units as required

Instances need
120 seconds to warm up before including in metric

Scale in
Enabled

Task 5: Load Testing

Step 1: Install the ‘stress’ package.

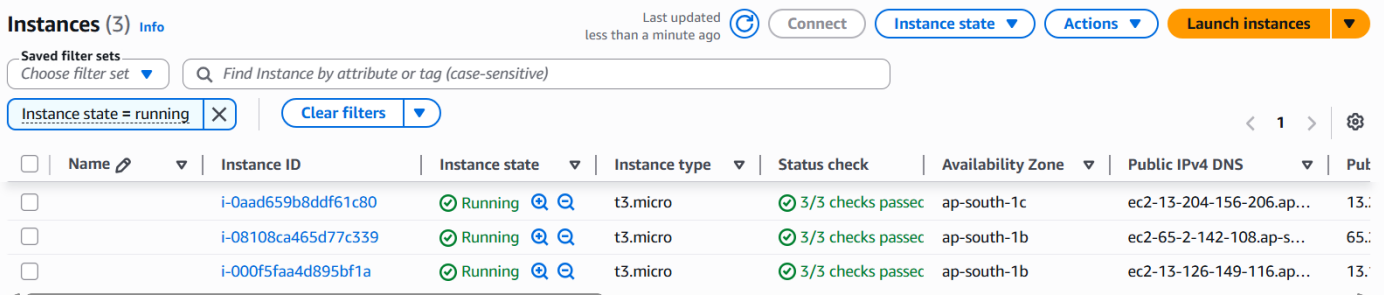
```
        _/m/'
[ec2-user@ip-172-31-27-24 ~]$ sudo yum install -y stress
Last metadata expiration check: 0:17:37 ago on Tue Jun  9 14:57:54 2026.
Dependencies resolved.
================================================================================================================================================
 Package                         Architecture                    Version                                     Repository                             Size
================================================================================================================================================
Installing:
 stress                          x86_64                          1.0.7-2.amzn2023.0.1                        amazonlinux                            34 k

Transaction Summary
================================================================================================================================================
Install  1 Package

Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm                                                                    526 kB/s |  34 kB     00:00
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                     338 kB/s |  34 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                            1/1
  Installing       : stress-1.0.7-2.amzn2023.0.1.x86_64                                                                                         1/1
  Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64                                                                                         1/1
  Verifying        : stress-1.0.7-2.amzn2023.0.1.x86_64                                                                                         1/1
```

Step 3: Enter the command #stress --cpu 4 --timeout 600

After a few minutes, the Auto Scaling Group scales out and increases the number of instances from 2 to 3.



Instances (3) Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set | Find Instance by attribute or tag (case-sensitive)

Instance state = running | Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Pub
	i-0aad659b8ddf61c80	Running	t3.micro	3/3 checks passed	ap-south-1c	ec2-13-204-156-206.ap...	13.
	i-08108ca465d77c339	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-65-2-142-108.ap-s...	65.
	i-000f5faa4d895bf1a	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-13-126-149-116.ap...	13.

Task 6: Auto Healing Test

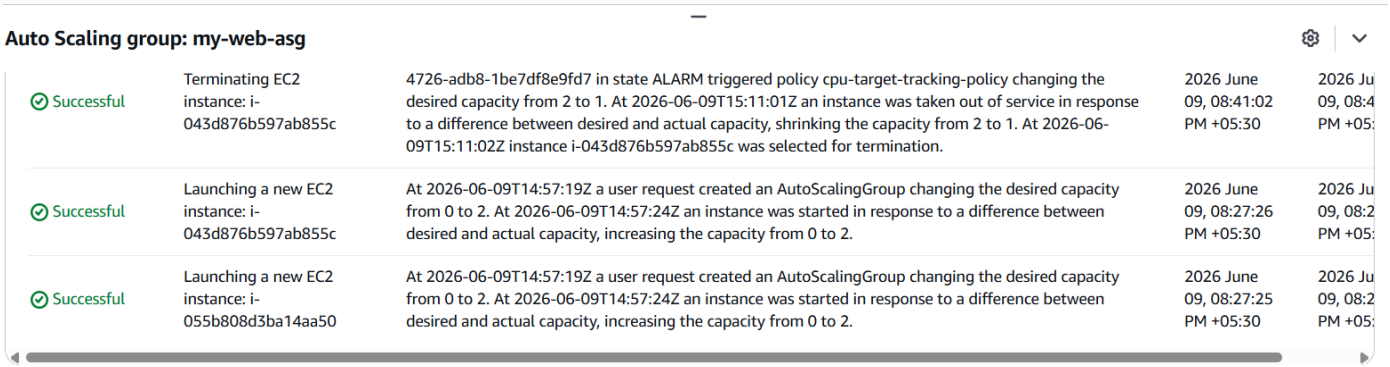
Step 1: Terminate one instance.

Step 2: The Auto Scaling Group automatically launches a replacement instance to maintain the desired capacity.

Successful	Launching a new EC2 instance: i-0aed8842987b89168	At 2026-06-09T15:31:29Z an instance was launched in response to an unhealthy instance needing to be replaced.

Task 7: Scale In and Cost Optimization

Stop the load-generation command and wait 10–15 minutes. As CPU utilization decreases, the Auto Scaling Group automatically scales in and reduces the number of running instances.



Auto Scaling group: my-web-asg

Successful	Terminating EC2 instance: i-043d876b597ab855c	4726-adb8-1be7df8e9fd7 in state ALARM triggered policy cpu-target-tracking-policy changing the desired capacity from 2 to 1. At 2026-06-09T15:11:01Z an instance was taken out of service in response to a difference between desired and actual capacity, shrinking the capacity from 2 to 1. At 2026-06-09T15:11:02Z instance i-043d876b597ab855c was selected for termination.	2026 June 09, 08:41:02 PM +05:30	2026 Ju 09, 08:4 PM +05:
Successful	Launching a new EC2 instance: i-043d876b597ab855c	At 2026-06-09T14:57:19Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-06-09T14:57:24Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 June 09, 08:27:26 PM +05:30	2026 Ju 09, 08:2 PM +05:
Successful	Launching a new EC2 instance: i-055b808d3ba14aa50	At 2026-06-09T14:57:19Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-06-09T14:57:24Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 June 09, 08:27:25 PM +05:30	2026 Ju 09, 08:2 PM +05: